Digital VLSI Design — Course Project

ML-Driven PVT Variation Aware Leakage Estimation

1 Overview

In this project, you will estimate the leakage current of CMOS circuits under process, voltage and temperature (PVT) variation using machine learning instead of full SPICE simulation. Using your assigned model file, first simulate single NMOS and PMOS transistors in their ON and OFF states. Use that data to train four ML models that predict each terminal's currents (I_d, I_g, I_s, I_b). Next, use only these four models to predict the leakage of INV, NAND2, NOR2, AND2 and OR2 for every input combination, and check each prediction against a SPICE simulation of the gate's transistor netlist. Finally, apply the same method to a larger circuit, either a **3-bit carry-lookahead adder** or an **8-bit parity checker**. Build its leakage from your gate-level predictions and compare it with a SPICE simulation of the complete netlist. Your target is upto **99% mean accuracy**.

2 Your assignment

Each team is assigned a model file, a nominal supply voltage and a test circuit by the TAs. Use only those as assigned to your team.

Item	**Your value**
Model file	As assigned (`16nm_HP`, `16nm_LP`, `22nm_HP`, `22nm_LP`, `22nm_MGK`, `32nm_HP`, `32nm_LP`, `32nm_MGK`, `45nm_HP`, `45nm_LP`, `45nm_MGK`)
Nominal VDD	As assigned (0.9 V or 1 V)
Channel length L	Feature size of your node (e.g. 16 nm for a 16 nm file)
Widths	NMOS and PMOS: $W = 1 \times L_{\min}$ to $8 \times L_{\min}$
Test circuit	As assigned (3-bit CLA or 8-bit Parity Checker)

3 Workflow

Part A: Transistor models

1. **Generate data.** For each of the four cases (NMOS ON, NMOS OFF, PMOS ON, PMOS OFF), generate an initial set of 10,000 random samples. For each sample, independently draw all input parameters specified in Section 4, including process parameters (P), supply voltage (V_{DD}), temperature (T), transistor width (W), and terminal voltages (V_d, V_g, V_s, V_b). Apply each sample to the corresponding single-FET SPICE netlist and simulate the transistor to obtain its terminal currents (I_d, I_g, I_s, I_b). Store the sampled input parameters and their corresponding simulated terminal currents as one dataset for each case. The number of samples may be increased or decreased as needed. The primary objective is to achieve high prediction accuracy on unseen data, preferably with the smallest dataset possible without compromising accuracy.
2. **Train models.** Randomly split each dataset into training (70 %, 7,000 samples), validation (15 %, 1,500 samples) and test (15 %, 1,500 samples) sets, using a fixed random seed. Alternatively, an 80:10:10 split may be used. Train and compare different ML models for each case, select the best-performing model on the validation set, and report its performance on the test set. The primary objective is to achieve the highest possible prediction accuracy on unseen data.

Part B: Gate-level validation

3. **SPICE reference.** Simulate every gate in Section 5 as a transistor-level netlist, for every input combination at 200 random corners(200 * 4 i.e for 00,01,10,11). These results are for evaluation only and must not be used for training.
4. **ML prediction.** For the same gates, input combinations and process corners used in Step 3, use the pre-recorded internal node voltages of the PMOS pull-up stack and NMOS pull-down stack, along with the output voltage, obtained from the SPICE simulations. Use these voltages to determine the terminal voltages (V_d, V_g, V_s, V_b) of each transistor in the gate. Feed the corresponding terminal voltages and other required parameters into your trained transistor models to predict the terminal currents, and use these predicted currents to calculate the logic gate leakage.
5. **Compare.** Report the accuracy of the ML prediction against SPICE for every gate and input combination.

Part C: Test-circuit validation

6. **SPICE reference.** Simulate your assigned test circuit as a complete transistor-level netlist, for every input combination at the same 200 corners used in Part B and obtain the circuit overall leakage current.
7. **ML prediction.** Build the test circuit from your gate models of Part B, which in turn call the transistor models. For each input vector, determine the logic value at the inputs of every gate, call the corresponding gate model, and sum the logic gate leakages to get the overall leakage current.
8. **Compare.** Report the accuracy of the hierarchical ML prediction against the SPICE netlist, and submit the deliverables in Section 6.

4 Parameters to sweep

Each of the 10,000 samples per case is generated by drawing every parameter below independently at random. Draw VDD first, since the ranges of V_d and V_s depend on it.

Parameter	**Range or distribution**
VDD	Uniform, assigned VDD ±10 %
Drain voltage V_d	Uniform, 0 to VDD
Source voltage V_s	Uniform, 0 to VDD
Width W	Uniform, $1 \times L_{\min}$ to $8 \times L_{\min}$
Temperature	Uniform, −40 to 125 °C
`vth0`, `toxe`, `ndep`, `u0`, `xj`	Uniform, nominal value from your model file ±10 %. Separate values for NMOS and PMOS

The gate and body voltages are fixed by the case:

Case	Gate voltage	Body voltage
NMOS ON	VDD	0
NMOS OFF	0	0
PMOS ON	0	VDD
PMOS OFF	VDD	VDD

Netlist requirements

Every netlist (transistor, gate and test circuit) must include `.options gmin=1e-18 abstol=1e-18 reltol=1e-6 vntol=1e-9`, and in the single-transistor netlists every terminal must have its own voltage source.

5 Gates and test circuit

Gates

Use standard static CMOS gates. AND2 and OR2 are a NAND2 or NOR2 followed by an INV.

Gate	Inputs	Internal node
INV	A = 0, 1	None
NAND2	AB = 00, 01, 10, 11	x (between the series NMOS)
NOR2	AB = 00, 01, 10, 11	y (between the series PMOS)
AND2	AB = 00, 01, 10, 11	x and the NAND2 output
OR2	AB = 00, 01, 10, 11	y and the NOR2 output

This gives 18 cases, each at 200 corners, in both SPICE and ML.

A *corner* is one random draw of VDD, temperature and process parameters, using the distributions in Section 4. All NMOS in a gate share one draw and all PMOS share another. Gate inputs are applied at exactly 0 or VDD.

Simulate AND2 and OR2 in SPICE as complete gates; do not add separately simulated NAND2 and INV results.

For every case, record the SPICE leakage, the ML leakage, the internal node voltages from both, and the output voltage from SPICE.

Test circuit

Simulate the assigned test circuit in SPICE as one complete netlist, at the same 200 corners and for every input combination. For the ML prediction, compose the circuit from your gate models and sum the gate leakages. This assumes every gate input sits at exactly 0 or VDD and ignores loading between gates, so expect the accuracy to be somewhat lower than at the gate level.

6 Deliverables

1. **Datasets:** four CSV files (NMOS ON, NMOS OFF, PMOS ON, PMOS OFF), 10,000 rows each, with columns `vdd, temp, w, vd, vs, vg, vb, vth0, toxe, ndep, u0, xj, id, ig,`

`is, ib`.

2. **Example netlists:** one single-transistor netlist per case, one SPICE netlist per gate, and the netlist of your test circuit.
3. **Data check:** the Kirchhoff check for each case (sum of the four terminal currents) and three trend plots (OFF current vs temperature, OFF current vs `vth0`, gate current vs `toxe`).
4. **Model comparison table:** 4 model types $\times$ 4 cases, with R^2, median error and 95th-percentile error on the validation set.
5. **Chosen models:** the test-set median and 95th-percentile error of each, a predicted-vs-SPICE plot for each case, and the four saved model files.
6. **Gate results table:** for each of the 18 gate and input cases, the SPICE median, ML median, bias, median error and 95th-percentile error.
7. **Gate parity plot:** ML leakage against SPICE leakage for all gate simulations, on log axes.
8. **Test-circuit results:** the same metrics as the gate results table for your test circuit, and a parity plot of ML against SPICE leakage on log axes.
9. **Code:** scripts that regenerate everything above from the model file, run in order, with fixed random seeds.
10. **Report:** a typed report covering the following.
    - **Setup:** your assigned model file, nominal VDD, test circuit, and the simulator and version used.
    - **Data generation:** how the samples were drawn, how the netlists were built, how the data was split into training, validation and test sets, and the results of the data check.
    - **Models:** the four model types tried, their main hyperparameters and any preprocessing (e.g. predicting $\log|I|$ instead of I), the model comparison table, and the reason for each final choice.
    - **Gate results:** the gate results table and parity plot, with a comment on which gates and input combinations are most and least accurate.
    - **Test-circuit results:** the test-circuit metrics and parity plot, compared with the gate-level accuracy.
    - **Error analysis:** where the largest errors occur (which case, gate, input combination or corner conditions, e.g. high temperature or low `vth0`), and your explanation of why.
    - **Limitations and improvements:** what limits the accuracy of your approach, and what you would change to improve it.